

Project Title: Employee Attrition and Workforce Analytics Using MySQL Workbench

Introduction

This project analyzes employee attrition patterns using SQL on the IBM HR Analytics dataset. The objective is to identify key factors associated with employee turnover, including department, overtime, salary levels, and tenure. Understanding these patterns can help organizations design better retention strategies and workforce planning policies.

Dataset Description

This project analyzes employee attrition patterns using SQL on the IBM HR Analytics dataset. The objective is to identify key factors associated with employee turnover, including department, overtime, salary levels, and tenure. Understanding these patterns can help organizations design better retention strategies and workforce planning policies. (The dataset was imported into MySQL and analyzed using SQL queries.)

Questions

1. What is the overall employee attrition rate in the organization?
2. Which departments experience the highest employee turnover?
3. Are employees who work overtime more likely to leave the company?
4. How does salary vary across departments and job roles?
5. Are employees with shorter tenure more likely to leave?
6. Which employees earn more than the company-wide average salary?

Methodology

The dataset was imported into MySQL Workbench. SQL queries were used to perform exploratory data analysis, including aggregations, grouping, filtering, and subqueries. The analysis focused on identifying patterns in attrition across departments, overtime status, salary distribution, and employee tenure.

Key Findings

1. Overall Attrition Trends - The organization has an attrition rate of 16.12%, indicating a moderate but significant level of employee turnover. Out of 1,470 employees, 237 have left, suggesting that retention is a meaningful concern for the organization.
2. **Department-wise Attrition Patterns** - Attrition is concentrated in specific departments:
 - Research & Development → 133 employees
 - Sales → 92 employees

- Human Resources → 12 employees

Observation:

The highest attrition occurs in Research & Development, which is a core functional area. This suggests that operational roles are more vulnerable to turnover compared to support functions like HR.

3. Compensation Disparities and Attrition

Average monthly income across departments shows:

- Sales → Highest (approx. 6959)
- HR → Moderate (approx. 6654)
- R&D → Lowest (approx. 6281)

Additionally:

Employees who left → Avg salary ~4787

Employees who stayed → Avg salary ~6832

Observation:

Employees who leave earn significantly lower salaries than those who stay. At the same time, R&D, despite having the highest attrition, has the lowest average salary among major departments. This indicates a strong relationship between compensation and attrition.

4. Job Role Hierarchy and Pay Structure

Salary distribution across job roles shows:

- Highest → Manager (approx. 17181), Research Director (approx. 16033)
- Lowest → Sales Representative (approx. 2626), Laboratory Technician (approx. 3237)

Observation:

There is a steep hierarchical pay structure. Entry-level roles earn substantially less than managerial positions, which may contribute to dissatisfaction and higher turnover among junior employees.

5. Tenure and Early Attrition Risk

Attrition is highest among employees with shorter tenure:

- Year 1 → 59 employees left
- Year 2 → 27
- Year 3 → 20

Declines significantly from year 4 onwards.

Observation:

Attrition is heavily concentrated in the first 1–3 years of employment, indicating early-stage disengagement, poor role fit, or unmet expectations.

6. Impact of Overtime on Attrition

Employees working overtime show higher attrition, suggesting that workload and work-life imbalance may be contributing factors to turnover.

7. Promotion Patterns and Career Stagnation - Employees with the longest time since last promotion show up to 15 years without promotion**Observation:**

A subset of employees has experienced prolonged periods without promotion, indicating potential career stagnation, which can negatively impact motivation and retention.

8. Income Distribution and Organizational Inequality - Income distribution appears to be uneven, with higher earnings concentrated among senior roles. This reinforces organisational hierarchy and may contribute to dissatisfaction among lower-paid employees.**Further Analysis**

- High attrition in R&D combined with lower average salaries suggests a mismatch between workload intensity and compensation.
- Early-career attrition indicates irregularities in onboarding, role alignment, or employee expectations.
- Overtime-driven attrition highlights the importance of work-life balance policies.
- Salary inequality across job roles reflects a steep hierarchical structure, impacting morale significantly.

My Recommendations**1. Review Compensation Structures in high-turnover departments.**

- Reassess salary levels, especially in R&D and entry-level roles.
- Introduce competitive compensation to reduce turnover among critical employees.

2. Improve Early Employee Engagement

- Strengthen the structure of onboarding processes.

- Introduce mentorship and structured career guidance in the first 1–3 years, as well as retention programs.

3. Address Workload and Overtime

- Monitor and regulate overtime.
- Promote work-life balance initiatives to reduce burnout.

4. Strengthen Career Progression Pathways

- Reduce stagnation by ensuring regular promotion reviews.
- Introduce transparent growth pathways and skill development programs.

5. Focus on Role-Specific Retention Strategies

- Develop targeted retention strategies for high-attrition roles (e.g., R&D staff, Sales executives).

6. Reduce Pay Inequality Perception

- Introduce performance-based incentives.
- Ensure fair compensation progression across roles.

Conclusion

An analysis of this dataset demonstrates that employee attrition is a multi-dimensional issue influenced by compensation, workload, tenure, and career progression. Addressing these factors through targeted organisational strategies can significantly improve retention and workforce stability.